

How long do you need?

Ground-Based Follow-up of LSST Transients — Hands-on Assignment

CASSA TDMMA Workshop 2026
Independent University, Bangladesh
Instrument: CRITO driving a simulated observatory
300 mm, 5.2 μ m, INDI simulators

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1 Setup

1.1 What you are flying

You will drive a complete observatory control system — CRITO — against *simulated* hardware. Every step is the real code path: the same INDI drivers, the same plate solver, the same exposure planner, the same archive writer. Only the photons are fictional.

Role	Driver you connect to	What it presents
Mount	INDI Telescope Simulator	300 mm optics, slews and tracks
Imaging camera	INDI CCD Simulator	5.2 μm pixels, star field from gsc
Guide camera	INDI CCD Simulator (guide role)	a second chip for PHD2
Focuser + wheel	INDI simulators	motorised focus, L/R/G/B/Ha/OIII/SII
Roof	INDI Dome Simulator	opens and closes
Broker feed	ALeRCE (ZTF, LSST-style)	live poll, or cached

Optical specification

Focal length	300 mm
Pixel size	5.2 μm
Filters	L, R, G, B, Ha, OIII, SII
Sub-exposure cap	300 s

These are the values used both on paper and in the Exposure tab.

1.2 Participant setup (five minutes, at your seat)

1. Open CRITO. The instructor will provide the workshop link during the session. Open the given link in a modern web browser. No terminal, VPN, or SSH tunnel is required. Keep the CRITO browser tab open for the entire exercise.

Workshop link. Use the given link only. If the page does not open, or the browser reports a connection error, tell the instructor rather than changing the address.

2. Claim your telescope.

1. Open the given link in your browser.
2. Log in with your workshop account: `workshopN` / the workshop password given by the instructor.
3. **Confirm the simulation badge** in the top bar. If it is not there, stop: you are pointed at real hardware and your account will (correctly) refuse to move anything.
4. On **Locations**, find your assigned telescope (`workshop1` $\rightarrow$ **Simulator 1**, and so on) and click **Operate**. Other assigned telescopes are disabled; an occupied one says **Busy**. Your claim is exclusive while you work.
5. **Devices** $\rightarrow$ **Scan** $\rightarrow$ **Auto-detect & connect all**. Six roles should bind: mount, camera, guide camera, focuser, filter wheel, roof.
6. **Exposure** tab $\rightarrow$ confirm a calibration table is loaded (gains 0/60/120/200; filters L, R, G, B, Ha, OIII, SII). If it says *no calibration table*, tell the workshop facilitator — see Section ??.

What your account can and cannot do. The workshop role is ranked with `viewer` and promoted to operator-level *only on a backend that declares itself a simulator*. So the same login flies your simulator and is a spectator on the real telescope. Reads — telemetry, archive, atlas, candidates — are never blocked, because watching the real rig work is half the point. A refused command names its reason:

```
{"detail":"this account is limited to a simulation backend
  - this one drives the real observatory"} [403]
```

1.3 Workshop setup (before the session)

1. Bring up the simulated observatory.

```
docker compose --env-file .env -f deploy/docker-compose.yml \
  --profile sim up -d --build
```

Run this command from the CRITO repository root. It starts one CRITO backend and one simulator container holding five isolated INDI/PHD2 pairs. Participants open the given link; the backend keeps all five telescope connections live and routes each authenticated workspace to its claimed rig. After changing `.env`, add `-force-recreate` to reload the container environment.

The star field is not optional. INDI's CCD simulator draws its stars by shelling out to `gsc`, the Guide Star Catalogue tool. Without it *every frame is blank noise*, and blank frames cannot be plate-solved — so centring, autofocus and polar alignment all fail in ways that look like solver bugs and are not. The image compiles `gsc` from source for exactly this reason. Sanity check:

```
docker compose --env-file .env -f deploy/docker-compose.yml exec indi-sim \
  gsc -c 83.822 -5.391 -r 30 -m 0 12 -n 5
```

Stars listed = fine. `gsc:` not found = wrong image.

2. Verify the simulated optics. The application image includes the ASTAP CLI and D05 catalogue. Compose sets all five simulators to 300 mm, matching `deploy/observatory.sim.yaml`. To verify simulator 1:

```
docker compose --env-file .env -f deploy/docker-compose.yml exec indi-sim \
  indi_getprop "Telescope Simulator.TELESCOPE_INFO.*"
```

Focal length	Pixels	Field (1024 px)	ASTAP database
300 mm	5.2 μ m	$\sim 1.0^\circ$	D05, ~ 100 MB

3. Backend environment.

```
# .env (one shared backend)
CRITO_SOLVE_SCIENCE_FRAMES=true          # REQUIRED for Part 2: writes WCS into lights
CRITO_CALIBRATION_FILE=calibration/minicam8.example.yaml
CRITO_AUTH_SECRET=<one long random workshop secret>
CRITO_ADMIN_PASSWORD=<strong instructor password>
CRITO_WORKSHOP_PASSWORD=<shared participant password>
CRITO_ALERCE_PROBABILITY=0.2              # loosen so the candidate list is never empty
CRITO_ALERCE_LOOKBACK_DAYS=14
```

The `-env-file .env` option is required so Compose uses these values when it creates the simulator service. Compose also supplies the simulation flag, telescope endpoints, 300 mm/5.2 μ m solver geometry, ASTAP path and D05 database path inside the containers. Do not put host macOS paths into these settings.

CRITO_SOLVE_SCIENCE_FRAMES is off by default. Light frames only carry CRVAL1/CRVAL2 (the plate-solved sky coordinates of the frame centre) when this flag is on — it costs a solve per frame, so the default is `false`. Part 2 asks participants to read those keywords out of the header. Turn it on, or drop that item from the hand-in.

4. One backend, five telescope leases. The telescope pool is declared in `deploy/observatory.sim.yaml`. It maps `workshop1` to `sim-1`, through `workshop5` to `sim-5`; operators may use any simulator and an administrator can force-release one. The browser renews an active claim every 30 seconds. An abandoned claim expires after 120 seconds, while a second user trying to claim the same telescope gets HTTP 409 instead of sharing its controls.

5. Accounts.

```
docker compose --env-file .env -f deploy/docker-compose.yml exec crito-sim \
  python scripts/seed_workshop_users.py --list
```

The backend provisions missing workshop accounts automatically from `CRITO_WORKSHOP_PASSWORD`. Existing accounts retain their stored passwords when the container restarts, so change them from the administrator account rather than expecting `.env` to overwrite them. New workshop accounts are **simulation-only by default**; the toggle to real hardware lives in the console's **Users** page, *Hardware* column, and is re-read from the database on every write (so revoking access takes effect immediately, not at the participant's next login).

1.4 Pre-flight checklist

Check	If it fails
☐ SIMULATION badge in the top bar	wrong backend — do not proceed
☐ six devices connected after <i>Scan</i>	re-run auto-detect; check <code>indiserver</code> is up
☐ a test frame shows <i>stars</i> , not noise	<code>gsc</code> missing from the sim image
☐ Exposure tab lists gains and filters	<code>CRITO_CALIBRATION_FILE</code> unset
☐ Candidates tab has ≥ 1 target	lower <code>CRITO_ALERCE_PROBABILITY</code> , re-poll
☐ a plate solve succeeds	FOV hint vs. database size mismatch

2 The physics you need

Everything below is in **electrons**. This is the whole of the reference sheet; Part 1 needs nothing else.

2.1 Geometry

$$\text{plate scale: } s = 206.265 \times \frac{p [\mu\text{m}]}{f [\text{mm}]} \quad [\text{arcsec/px}] \quad (1)$$

$$\text{PSF width: } \theta_{\text{px}} = \frac{\text{seeing} [\text{arcsec}]}{s}, \quad \sigma = \frac{\theta_{\text{px}}}{2.3548} \quad (2)$$

$$\text{aperture: } n_{\text{pix}} = \pi (1.5 \theta_{\text{px}})^2 \quad (3)$$

A radius of $1.5 \times \text{FWHM}$ encloses about 99 % of a Gaussian PSF's flux — that is the aperture the photometry is done in, and n_{pix} is how many pixels' worth of sky, dark and read noise you are forced to swallow along with the star.

2.2 Rates

$$\text{source: } S = 10^{-0.4(m-\text{ZP})} \quad [\text{e}^- \text{ s}^{-1}] \quad (4)$$

$$\text{PSF peak: } P = \frac{S}{2\pi\sigma^2} \quad [\text{e}^- \text{ s}^{-1} \text{ px}^{-1}] \quad (5)$$

The zero point ZP is defined as the magnitude that yields exactly $1 \text{ e}^- \text{ s}^{-1}$ through the whole system — optics, filter, sensor. It is measured per filter, on-sky, at your site; it is *not* a property of the camera.

2.3 The CCD equation

For a single sub-exposure of length t :

$$\text{SNR}_{\text{sub}} = \frac{St}{\sqrt{\underbrace{St}_{\text{source shot}} + n_{\text{pix}}(\underbrace{Bt}_{\text{sky shot}} + \underbrace{Dt}_{\text{dark}} + \underbrace{R^2}_{\text{read}})}} \quad (6)$$

with B the sky rate ($\text{e}^- \text{ s}^{-1} \text{ px}^{-1}$), D the dark current ($\text{e}^- \text{ s}^{-1} \text{ px}^{-1}$) and R the read noise (e^- , *not* squared in the table — square it here).

2.4 The two bounds on a sub-exposure

$$\text{sky-limited floor: } t \geq \frac{k R^2}{B}, \quad k = 10 \quad (7)$$

$$\text{saturation ceiling: } t \leq \frac{0.7 \text{FW}}{P_{\text{brightest}} + B + D} \quad (8)$$

Equation (7) says: make the sub long enough that sky shot variance is at least ten times the read variance, otherwise you pay the read-noise toll on every single frame you stack. Equation (8) says: the brightest star you care about must stay in the linear part of the well — 70 % of full well is the usual safe fraction. A third bound is operational, not physical: **your mount and guider cap the sub** regardless of what the photons allow.

2.5 Stacking

$$\text{SNR}_{\text{total}} = \text{SNR}_{\text{sub}} \sqrt{N}, \quad N = \left\lceil \left(\frac{\text{SNR}_{\text{required}}}{\text{SNR}_{\text{sub}}} \right)^2 \right\rceil, \quad t_{\text{total}} = N t \quad (9)$$

$$\sigma_{\text{mag}} = \frac{1.0857}{\text{SNR}_{\text{total}}} \quad (10)$$

2.6 The calibration table

These are the constants the planner is loaded with (`calibration/minicam8.example.yaml`). Read noise and full well change with gain, because on a CMOS sensor they must; sky rate and zero point are per *filter* and per *site*, and are gain-independent.

This exercise uses the gain-60 row, and every filter row you decide to survey. Sky rate and zero point are properties of the filter and the site, not of the camera — which is exactly why the choice of band changes the answer.

Gain setting	Read noise R [e ⁻]	Full well FW [e ⁻]	System gain [e ⁻ /ADU]
0	5.0	51 000	0.80
60	3.2	38 000	0.45
120	1.1	13 000	0.16
200	0.9	9 000	0.10

Filter	Sky B [e ⁻ s ⁻¹ px ⁻¹]	Zero point ZP
L	8.00	21.0
R	3.00	20.3
G	3.50	20.4
B	2.00	19.8
Ha	0.12	18.6
OIII	0.15	18.8
SII	0.10	18.3
Other		
Pixel pitch p		5.2 μm
Dark current D at -10 °C	0.01 e ⁻ s ⁻¹ px ⁻¹	
Focal length f		300 mm

3 How to think physically and choose an exposure

The equations above are compact, but operating a telescope requires a physical picture of what they mean. The central distinction is:

The rule to remember. Sub-exposure length controls how efficiently and safely the detector is read. Total integration time controls how deep the observation goes.

Once a sub is long enough that read noise is unimportant, making that sub still longer gives little extra statistical efficiency. More total time still helps, but it can be collected as several moderate subs instead of one fragile long one.

3.1 The pixel as a bucket

Imagine every pixel as a bucket left outside in the rain. During an exposure, electrons accumulate in it from several physically different sources:

Quantity	Physical picture	Consequence
Source S	Rain from the astronomical object that we want.	More source electrons improve the measurement, but their Poisson fluctuation is unavoidable.
Sky B	Unwanted rain falling into every pixel in the aperture.	A brighter sky raises the noise even after its mean level is subtracted.
Dark current D	A slow leak into the bucket caused by the detector itself.	Cooling suppresses it; subtracting a dark frame removes its mean pattern, not its shot noise.
Read noise R	Uncertainty added when the bucket is measured and emptied.	It is paid once per pixel <i>per frame</i> , not once per second.
Full well FW	The finite capacity of the bucket.	Near the top the response becomes nonlinear; at the top the pixel saturates and brightness information is lost.

A short exposure empties the bucket frequently. This protects bright stars and limits the time lost to a bad frame, but every read introduces another R^2 . A long exposure pays read noise less often, but gives saturation, guiding, clouds, wind and aircraft more time to damage a larger piece of the observation. The useful exposure window lies between those two failure modes.

3.2 There are two clocks, not one

Observers often ask “what is the exposure time?” as if it were one number. There are really two:

1. The **sub-exposure time** t : how long the shutter stays open before one readout.
2. The **total integration** $T = Nt$: how many source photons are accumulated across the complete stack.

For a fixed total time T , with $N = T/t$ equal subs, Equation (6) can be written as

$$\text{SNR}(T, t) = \frac{ST}{\sqrt{ST + n_{\text{pix}} \left[(B + D)T + \frac{T}{t} R^2 \right]}}. \quad (11)$$

This form contains the main exposure-planning insight. At fixed T , the source, sky and dark terms do not care how the time is divided. Only the read term, $(T/t)R^2$, depends on the number of frames. Increasing t helps strongly while that term matters. Once it is small beside the sky term, increasing t has very little left to improve.

What longer subs cannot do. A 300-s sub does not collect photons faster than three 100-s subs. Both contain 300 seconds of source and sky. The single long exposure merely pays read noise once instead of three times. If 100-s subs are already sky-limited, that saving is small.

3.3 The lower bound: when is a sub too short?

The sky contributes variance Bt per pixel, while one read contributes variance R^2 . CRITO asks for the sky variance to be ten times the read variance:

$$q_{\text{sky}} \equiv \frac{Bt}{R^2} \geq 10, \quad t_{\min} = \frac{10R^2}{B}. \quad (12)$$

The dimensionless number q_{sky} is a useful diagnostic:

q_{sky}	Interpretation
< 1	Strongly read-noise limited. Longer subs can improve efficiency substantially.
$1\text{--}10$	Transition region. The frames are usable, but stacking pays a noticeable read-noise penalty.
≥ 10	Comfortably sky-limited by the planner's convention. Longer subs have diminishing statistical benefit.

For the workshop R-band example, $R = 3.2\text{ e}^-$ and $B = 3.0\text{ e}^- \text{ s}^{-1} \text{ px}^{-1}$, so

$$t_{\min} = \frac{10(3.2)^2}{3.0} = 34.1\text{ s}. \quad (13)$$

A 5-s frame is far below this floor and repeatedly pays the amplifier's uncertainty. A 34-s frame has just reached the adopted threshold. A 100-s frame is comfortably past it.

The factor ten is a policy, not a law of nature. At $q_{\text{sky}} = 10$, adding read noise increases the sky-plus-read RMS by $\sqrt{1 + 1/10} - 1 \simeq 4.9\%$. Choosing a different acceptable penalty would move the floor, but not change the physics.

3.4 The upper bounds: when is a sub too long?

There is not one universal upper bound. The usable ceiling is

$$t_{\max, \text{usable}} = \min(t_{\text{saturation}}, t_{\text{guiding}}, t_{\text{cadence}}, t_{\text{operations}}). \quad (14)$$

Saturation. The brightest relevant star sets the peak-pixel rate. CRITO protects 70 % of full well using Equation (8). Saturation is not merely an ugly white pixel: once the well is nonlinear or full, the original flux cannot be reconstructed reliably. Enter the magnitude of the brightest star whose photometry matters, which is often not the transient itself.

Guiding and tracking. A mount may keep an acceptable PSF for 100 seconds but trail at 300 seconds. Periodic error, polar drift, wind and differential flexure impose a ceiling even when the detector has plenty of headroom. This cap should come from measured stellar elongation or guiding performance, not from optimism.

Science cadence. An exposure averages everything that happens during it. A 600-s exposure cannot resolve a 60-s change. For a rapidly changing transient, occultation or eclipse, time resolution may impose a much shorter ceiling than saturation or guiding.

Operational risk. Cosmic rays, satellites, clouds, vibrations and focus excursions can spoil a frame. Ten 60-s frames and one 600-s frame contain the same nominal integration, but rejecting one bad 60-s frame costs far less. Multiple frames are also needed for robust outlier rejection.

3.5 Short subs versus long subs

Neither is intrinsically superior. The correct choice depends on which cost is active.

	Shorter subs	Longer subs
Read noise	Paid more often; harmful below $t_{\min}$.	Paid fewer times; valuable when sky is faint.
Bright stars	More dynamic range and fewer saturated pixels.	Saturation and nonlinearity become more likely.
Guiding	Less trailing and less sensitivity to brief disturbances.	Requires stable tracking for the whole frame.
Frame rejection	A rejected frame loses little observing time.	One failure can discard many minutes.
Cadence	Better time resolution.	Averages rapid variability.
Overhead/data	More downloads, files and storage; lower duty cycle.	Fewer files and less relative readout overhead.

If the camera takes t_{read} seconds to read a frame, the simple observing efficiency is

$$\eta = \frac{t}{t + t_{\text{read}}}. \quad (15)$$

For a 2-s readout, 5-s subs have only 71 % open-shutter efficiency, while 100-s subs have 98 %. Dithering and settling add further overhead. This is a real argument against needlessly short subs even when read noise is tiny.

3.6 Why CRITO starts near three times the floor

When no explicit sub is requested, the planner uses the practical heuristic

$$t_{\text{recommended}} = \min(t_{\text{max,usable}}, 3t_{\min}), \quad (16)$$

provided the floor lies below the ceiling. Three times the floor is not a unique physical optimum. It is a robust operating point: clearly beyond the read-noise transition, but normally shorter and safer than the maximum possible exposure.

For the slide's R-band example,

$$t_{\min} = 34.1 \text{ s}, \quad t_{\text{guide}} = 300 \text{ s}, \quad t_{\text{recommended}} \simeq 3(34.1) = 102.3 \text{ s}. \quad (17)$$

Why not 34 seconds? That is only the threshold. Why not 300 seconds? Once 102-s frames are well sky-limited, the longer frame gains little read-noise efficiency while increasing saturation, guiding and frame-loss risk. CRITO then computes SNR_{sub} and uses Equation (9) to choose N ; in the example, 13 subs give about 22.2 minutes on sky and $\text{SNR} \simeq 5$.

3.7 How gain moves both walls

On a modern CMOS detector, increasing gain commonly lowers read noise and also lowers effective full well. Consequently,

$$R \downarrow \implies t_{\min} \downarrow, \quad (18)$$

$$\text{FW} \downarrow \implies t_{\max} \downarrow. \quad (19)$$

High gain can therefore make short subs efficient for a faint target, especially through narrowband filters, but it makes bright stars saturate sooner. Low gain usually retains more dynamic range but demands longer subs before read noise is buried.

If $t_{\min} > t_{\max}$, no exposure can be both sky-limited and unsaturated. This is a genuine physical conflict, not a calculator error. The observer can:

1. lower the gain to gain full-well capacity;
2. use shorter, mildly read-noise-limited subs and accept the calculable penalty;
3. tolerate saturation if the bright stars are scientifically irrelevant;
4. change the optical configuration; or
5. take separate short and long sequences for dynamic range.

CRITO chooses the upper ceiling in this conflict, protects the bright pixels, and reports a warning about the read-noise penalty.

3.8 How sky, filter and seeing change the answer

Bright urban broadband sky. A large B makes $t_{\min} = 10R^2/B$ short. The detector may become sky-limited in seconds. Very long broadband subs then accumulate proportionally more sky and risk saturation without providing much read-noise advantage.

Dark sky or narrowband. A narrowband filter suppresses most sky photons, making B small and $t_{\min}$ long. This is a sound physical reason for multi-minute subs: without them, read noise remains a significant fraction of every frame. The same filter also changes the zero point and therefore the source rate, so “darker sky” does not automatically mean “faster observation.”

Seeing. Poor seeing spreads the star across more pixels. The peak pixel fills more slowly, which can raise the saturation ceiling, but the aperture includes more sky pixels and faint-source SNR becomes worse. Good seeing concentrates the source into fewer pixels, improving faint-source SNR while making the central pixel saturate sooner. This is why seeing belongs in the exposure calculation.

3.9 A decision procedure at the telescope

Use this sequence rather than beginning with a familiar exposure such as “always 300 seconds”:

1. **State the measurement.** Choose the required SNR from the scientific use: detection, rough photometry, precise photometry, or time-series work.
2. **Choose a filter for the science.** Throughput matters, but continuity with an existing light curve may matter more than the fastest band.
3. **Choose gain.** Balance read noise against full well using the characterized gain table.
4. **Use current conditions.** Supply the actual seeing, sky rate and detector temperature where possible.
5. **Compute the floor.** Evaluate $t_{\min} = 10R^2/B$ and identify whether candidate subs are read-noise limited.
6. **Compute every ceiling.** Check saturation using the brightest relevant field star, then apply guiding and cadence caps.
7. **Choose inside the window.** Start near 1–3 times the floor, not automatically at the ceiling.
8. **Let total time follow from SNR.** Calculate SNR_{sub} , then N and $T = Nt$. If T does not fit the visibility window, change the scientific requirement or instrument choice; changing only the packaging of the same photons will not solve it.

9. **Take a test frame.** Check the measured background, brightest pixel, FWHM/HFR, ellipticity, star count and guiding. A plan is a prediction; the first exposure is evidence.
10. **Adjust one cause at a time.** Shorten for saturation, trailing or cadence. Lengthen when read noise or overhead dominates. Add more subs when the individual exposure is already in the safe sky-limited window.

Common wrong decisions.

- **“Longer is always deeper.”** Total time is deeper; one longer frame is mainly a different way of packaging that time.
- **“Stacking removes sky noise.”** Sky subtraction removes the mean sky level, not the Poisson fluctuations already recorded.
- **“The target is faint, so nothing will saturate.”** A bright field star may set the ceiling.
- **“Higher gain means greater sensitivity.”** It changes the electronic conversion, read noise and full well; it does not create photons.
- **“The planner’s recommendation is a command.”** It assumes the calibration and conditions are correct. The test frame remains part of the decision.

The final physical rule. Increase the sub length only until read noise and overhead stop mattering. After that, increase the number of safe subs rather than the risk carried by each individual frame.

4 The assignment

Part 1 — Plan it

35 minutes, on paper

The situation. It is 19:40 on a clear September evening in Dhaka. An alert comes through the broker: a supernova candidate discovered two nights ago, currently at $R = 20.0$ and still rising. The follow-up coordinator sends the same message to every node in the network:

“We need one photometric point on this object tonight, good to about a tenth of a magnitude, that we can put on the light curve. Which filter — and how long do you need?”

You have to answer both questions before you start — a band, and a number of minutes. The object sets at 01:20 and three other targets are queued behind it, so an open-ended exposure is not an option, and a total that is too short returns a point too noisy to use.

Part 1 is that estimate: **which band should this point be taken in, and how much telescope time does it cost?** Both halves are marked. The filter is not handed to you — choosing it *is* part of the answer, and the arithmetic is what makes the choice defensible.

What you know when the alert arrives		Where it comes from
Target magnitude	20.0	the broker's latest photometry
Source colour	take it as zero	a simplification: assume $m = 20.0$ in <i>every</i> band, so that the filter comparison is about the instrument and not about the star
Required SNR	10	the coordinator asked for ~ 0.1 mag; by Eq. (10), $\sigma_{\text{mag}} = 1.0857/\text{SNR}$
Filter	your choice	all seven rows of the filter table are available to you
Existing curve	six points in R	what the other nodes have already contributed over two nights
Gain setting	60	the rig's standard science gain
Sensor temperature	-10°C	what the cooler holds on a September night
Seeing	3.0 arcsec FWHM	an ordinary night at this site
Focal length	300 mm, pixels $5.2\ \mu\text{m}$	the instrument you are connected to
Sub-exposure cap	300 s	past this the guiding drifts and stars trail

No console. No calculator app you cannot explain. Derive, *in this order*:

1.1 Geometry: the plate scale s , the FWHM in pixels, and the aperture size n_{pix} . Do this *once*: none of the three contains a filter property. (n_{pix} is how many pixels' worth of sky you must measure in order to measure one star.)

1.2 The band survey. For **L**, **R**, **B** and *one* narrowband of your choosing, work out four rows:

- the **sky-limited floor** $t \geq 10R^2/B$ — the shortest sub worth taking; below it you pay the read-noise toll on every frame in the stack;
- a **sub length** inside the window the floor and the 300s guiding cap leave, saying which of the two binds. *For at least one of your four bands the floor lands above the cap* — that is not an arithmetic error, and you should say in one line what it means;
- SNR **per sub** from Equation (6);
- the **number of subs** N and the **total on-sky time**.

1.3 Choose your band, and defend it in one sentence. The band with the shortest total is not automatically the right answer, and your sentence should not be about photons.

1.4 The saturation ceiling $t \leq 0.7\text{FW}/(P + B + D)$ for the band you chose, taking the target itself as the brightest thing in the aperture. Say whether it binds, and by what factor.

1.5 The noise budget of your chosen band: all four terms of Equation (6) separately, each as a fraction of the total. That is what tells you what is limiting the observation — and it is not the same answer in every band you surveyed.

1.6 The achieved SNR and the resulting σ_{mag} . Compare it with the tenth of a magnitude that was requested.

1.7 Seeing. Without redoing the arithmetic from scratch, say what your total becomes on a 2 arcsec night rather than a 3 arcsec one, and why. One line of reasoning is enough if it is the right line.

► **Hand in:** the **four-row survey table** *with the working*; the band you chose and the sentence defending it; which bound binds in each band; the four noise fractions; σ_{mag} ; and your seeing answer. A number without its working scores nothing. Then complete the reply: “Yes — _____ minutes on sky, _____ subs of _____ s in _____, photometry to $\pm$ _____ mag.”

Estimate first. Before working through the arithmetic, write down two things: how many minutes you expect a mag-20 star to need on this telescope for a 0.1 mag point, and which of the seven filters you expect to be quickest. Compare both with your final answer. Most estimates are out by a factor of several, and the size and direction of the error is worth knowing — and the guess about the filter is wrong more often than the guess about the time.

Part 2 — Fly it

45 minutes, on the simulator

- 2.1 Log in to your **workshop** account and complete the pre-flight checklist.
- 2.2 Put Part 1 into the **Exposure** tab: focal length 300 mm, pixel size 5.2 μm , magnitude 20.0, required SNR 10, **the filter you chose**, gain 60, temperature -10 , seeing 3.0, max sub 300 s. **The planner must agree with your paper.** If it does not, one of you is wrong — find out which, and write down what the disagreement was.
- 2.3 Now change only the filter, and step through the other three bands you surveyed. The planner should reproduce your four rows. Any row it does not reproduce is the row to go back and check.
- 2.4 Go to **Candidates**, poll ALerCE, and pick an observable target. Note its class, peak altitude, airmass and moon separation, and say in one line why it is or is not worth a slot tonight.
- 2.5 **Approve** → **Queue**, then **Observing** → **Launch**. Watch the whole chain and name each stage as it happens: slew → plate-solve → centre → autofocus → guide → expose → archive.
- 2.6 Open the archived frame and read its FITS header.

► **Hand in:** from the header — OBJECT, FILTER, EXPTIME, GAIN, EGAIN, READNOIS, CRVAL1, CRVAL2; and from the archive row — HFR and the star count. Plus one screenshot of the execution monitor mid-run.

Where each number lives. OBJECT, FILTER, EXPTIME, GAIN, EGAIN and READNOIS are written into the FITS header at the edge, when the frame is authored. CRVAL1/CRVAL2 are injected *after* a successful plate solve, and only when CRITO_SOLVE_SCIENCE_FRAMES=true. HFR and star count are *not* header keywords at all: they are measured at ingest by ASTAP and stored on the archive row — read them in the console’s Archive list.

Deliverable and marking

One page: **the survey table, the numbers, the header, one screenshot.** Be ready to state, in two minutes, your band and why, your total time, and *which* of the three limits — sky floor, saturation, guiding — set your sub length.

Part	Marks	What earns them
1	55	The four-band survey, correct and with working (25); the band chosen, with a defence that is not about photons (12); which bound binds in each band — including the one whose floor sits above the cap (8); the four noise fractions and what they say is limiting (5); σ_{mag} and the completed reply (5).
2	45	Planner reconciled with paper across all four bands, any disagreement diagnosed (15); execution chain named stage by stage (10); header and archive values transcribed correctly (15); candidate justified in one line (5).

5 Hints

Read these only when you are stuck. They are ordered, and each gives away a little more than the last.

Part 1

H1.1. Do the geometry first and write down three numbers before you touch the CCD equation: s in arcsec/px, FWHM in px, and n_{pix} . Every later step uses n_{pix} , and it is the single easiest place to lose a factor of π .

H1.2. $206.265 \times 5.2/300$ is just under 3.6 arcsec/px, so at 3 arcsec seeing the PSF is narrower than one pixel. The instrument is under-sampled — that is a property of a wide-field rig, not an error — and it is why n_{pix} comes out at about five pixels.

Note also that $n_{\text{pix}} \propto \text{seeing}^2$, and that Equation (3) contains no camera property. The seeing, not the sensor, sets how many pixels of background are measured alongside the star. This matters again at steps 1.5 and 1.7.

H1.3. The survey is much less work than it looks. The geometry does not change between bands — s , FWHM and n_{pix} contain no filter property — so only B and ZP move, and both are read straight off the table. One column of arithmetic, four times.

H1.4. If you take the sub as $3\times$ the floor, then $Bt = 30R^2$ by construction, and n_{pix} is fixed — so the sky term comes out to the same number of electrons in every broadband filter. Anything that differs between those bands is therefore coming from the source rate, not the sky.

H1.5. In a narrowband filter, $10R^2/B$ lands above 300s. You cannot reach the sky-limited regime there on this mount: the guider takes the sub away from you before the sky has had time to swamp the read noise. Work out what that does to the read-noise share of the variance, and you have the answer to 1.2's last bullet.

H1.6. The shortest total is not the answer to 1.3. Ask what the point is *for*: it has to join six existing points and be interpretable by somebody who did not take it. That is a sentence about the network, not about the telescope.

H1.7. For the ceiling you need the PSF *peak* rate, not the total: the star's light is spread over the PSF, and only the central pixel is at risk of saturating. That is Equation (5), and σ there is in *pixels*.

H1.8. The ceiling for a mag-20 star comes out at thousands of seconds. That is not an error. Consider whether saturation is the constraint that actually limits your exposure at this magnitude.

H1.9. A defensible rule for choosing the sub: *three times the sky-limited floor*, clamped into the window. At $3\times$ the floor, read noise is about 3% of your noise budget — comfortably sky-limited, without making each sub so long that one satellite trail costs you a big fraction of the night.

H1.10. N must be a whole number, and you must round *up*. Rounding up means you overshoot the required SNR slightly — report the SNR you actually achieve, not the one you asked for.

Part 2

H2.1. If the planner disagrees with your paper by a large factor, check the focal length and pixel size in the form *first*: they must read 300 mm and 5.2 μm . Selecting a saved setup with different optics replaces both fields without warning, and every result downstream changes with them.

H2.2. If the planner disagrees by a few percent only, you are both right and the difference is rounding — in n_{pix} , or in where you rounded N . Say so explicitly; that is a correct answer, not a failure.

H2.3. Empty candidate list? The broker returned nothing, which is a real operational condition, not a bug. Lower CRITO_ALERCE_PROBABILITY, widen CRITO_ALERCE_LOOKBACK_DAYS, or clear the classifier and re-poll.

H2.4. Nothing launches? Check for a *manual override* badge in the Observing tab. Any manual mount or camera command pauses the queue by design — a human touching the telescope always wins. Clear the override and resume.

H2.5. To read the header without leaving the terminal:

```
curl -s localhost:8000/api/images/<id>/fits -o frame.fits
python -c "from astropy.io import fits; print(repr(fits.open('frame.fits')[0].header))"
```

6 Solutions

All values below were computed with `crito.transient.exposure` against `calibration/minicam8.example.yaml`, and agree with the console to the digits shown. Small differences in an answer — a percent or two — are almost always rounding in n_{pix} or in N , and should be accepted.

6.1 Part 1, step by step

1.1 Geometry.

$$s = \frac{206.265 \times 5.2}{300} = \mathbf{3.575} \text{ arcsec/px}$$

$$\theta_{\text{px}} = \frac{3.0}{3.575} = \mathbf{0.839} \text{ px}, \quad \sigma = \frac{0.839}{2.3548} = 0.356 \text{ px}$$

$$n_{\text{pix}} = \pi (1.5 \times 0.839)^2 = \pi \times 1.259^2 = \mathbf{4.98} \text{ px}$$

The aperture is five pixels. The seeing disc is narrower than one pixel, so the instrument is under-sampled; the consequence is that measuring one star brings in only five pixels' worth of sky, dark and read noise. *None of this changes with the filter* — which is what makes the survey in 1.2 cheap.

1.2 The band survey. Only two numbers move between bands: the sky rate B and the zero point ZP. Worked in full for R , and then repeated:

$$S = 10^{-0.4(20.0-20.3)} = 10^{0.12} = 1.318 \text{ e}^- \text{ s}^{-1}, \quad t_{\text{min}} = \frac{10R^2}{B} = \frac{102.4}{3.0} = 34.1 \text{ s}$$

$$t = 3t_{\text{min}} = 102.4 \text{ s} \in [34.1, 300], \quad \text{SNR}_{\text{sub}} = \frac{St}{\sqrt{St + n_{\text{pix}}(Bt + Dt + R^2)}} = \frac{135.0}{41.5} = 3.25$$

$$N = \lceil (10/3.25)^2 \rceil = 10, \quad t_{\text{total}} = 1024 \text{ s} = 17.1 \text{ min}$$

S in $\text{e}^- \text{ s}^{-1}$; floor and sub in s; total in min								
Filter	S	floor	sub	binds	SNR_{sub}	N	total	read
L	2.512	12.8	38.4	$3 \times \text{floor}$	2.35	19	12.2	3.0 %
G	1.445	29.3	87.8	$3 \times \text{floor}$	3.07	11	16.1	3.0 %
R	1.318	34.1	102.4	$3 \times \text{floor}$	3.25	10	17.1	3.0 %
B	0.832	51.2	153.6	$3 \times \text{floor}$	3.08	11	28.2	3.0 %
[O III]	0.331	682.7	300.0	guiding cap	5.04	4	20.0	13.1 %
H α	0.275	853.3	300.0	guiding cap	4.56	5	25.0	15.6 %
[S II]	0.209	1024.0	300.0	guiding cap	3.76	8	40.0	18.3 %

Three things in that table are worth naming out loud.

The sky term is the same in every broadband row. Choosing $t = 3 \times 10R^2/B$ makes $Bt = 30R^2$ regardless of B , and n_{pix} is fixed by the seeing — so the sky contributes $4.98 \times 30 \times 10.24 = 1529 \text{ e}^-$ of variance in L, G, R and B alike. Everything that separates those four rows comes from the source rate, i.e. from the zero point. A filter does not win by having a darker sky; it wins by passing more of the star's light per unit of sky it lets through.

In the narrowbands the floor sits above the cap. $10R^2/B$ is 683–1024s in the three narrowband filters — two to three times the 300s the guider will give you. The sub is therefore *not* long enough to swamp the read noise, and the read share jumps from 3% to 13–18% of the variance. This is the one regime in the whole assignment where a quieter gain setting would genuinely help: at gain 200, $R = 0.9\text{ e}^-$ instead of 3.2, and the read term all but disappears.

It is also the cleanest illustration of why the mount is a photometric instrument. The limit here is not the sky, not the sensor and not the star — it is how long the guider can hold a star still. A better mount would make a narrowband point on this rig meaningfully cheaper. A better camera would not.

The rows are not fair to a real supernova. The survey assumes a colour-neutral source, $m = 20.0$ in every band. A young supernova near peak is blue, so its B magnitude would be brighter than its R . Even a full 0.2 mag of blue — $B = 19.8$ — only lifts SNR_{sub} to 3.68 and cuts B to 8 subs, 20.5 min: still worse than R . The ordering in the table is robust to any plausible colour, which is why the simplification is safe to make. Participants who raise the colour point have seen something real, and should be told that it does not change the answer here *and why*.

L is fastest. R is correct.. L takes **12.2 minutes** against R's 17.1 — it is the broadest filter, passes the most light, and on photons alone it wins outright. G beats R too, at 16.1 minutes. If the question were “how do I get a tenth of a magnitude in the least time”, the answer would be L and the assignment would be over.

The answer is R , and the defence is not about photons: **the point has to join six existing R points**. An L measurement is not on any standard photometric system, cannot be calibrated against a catalogue in the usual way, and cannot be plotted on the network's light curve — so it is five minutes faster and scientifically worthless for the job that was asked. A node that returns the fastest number instead of the usable one has misunderstood what it is part of.

Accept any answer that chooses R for a reason about comparability, calibration, or the existing light curve. *Accept with credit* an answer that chooses R but notes that G costs one minute less and would be the right call if the network standardised on G . *Do not accept* L, however good the arithmetic — and this is worth saying plainly, because the arithmetic being right is exactly what makes the wrong choice tempting.

1.3 The choice, and the sentence that defends it.

1.4 Saturation ceiling (in R).

$$P = \frac{S}{2\pi\sigma^2} = \frac{1.318}{2\pi \times 0.356^2} = \frac{1.318}{0.798} = 1.652\text{ e}^- \text{ s}^{-1} \text{ px}^{-1}$$

$$t_{\text{max}} = \frac{0.7 \times 38\,000}{1.652 + 3.0 + 0.01} = \frac{26\,600}{4.662} = \mathbf{5705\text{ s}}$$

Which bound binds. The binding ceiling is the guiding cap, not saturation, by a factor of 19. At 20th magnitude on a sensor with a 38 ke^- well, the star could integrate for an hour and a half before the central pixel filled; the limit is that the mount cannot track that long without trailing. Participants who report “saturation limits us to 300s” have mis-read their own arithmetic. Those who quote the factor of 19 have read it correctly — and those who noticed that the answer is *the mount* in all seven rows of the survey have read it better still.

1.5 The noise budget. At $t = 102.4$ s in R :

$$\begin{aligned}
 \text{signal} &= S t = 1.318 \times 102.4 = 135.0 \text{ e}^- \\
 \text{source shot} &= 135.0 \\
 \text{sky} &= n_{\text{pix}} B t = 4.98 \times 3.0 \times 102.4 = 1528.9 \\
 \text{dark} &= n_{\text{pix}} D t = 4.98 \times 0.01 \times 102.4 = 5.1 \\
 \text{read} &= n_{\text{pix}} R^2 = 4.98 \times 10.24 = 51.0 \\
 \text{total variance} &= 1720.0 \text{ e}^{-2}, \quad \text{noise} = 41.5 \text{ e}^- \\
 \text{SNR}_{\text{sub}} &= \frac{135.0}{41.5} = \mathbf{3.26}
 \end{aligned}$$

The frame is **sky-limited**: sky is 88.9 % of the variance, the source itself 7.8 %, read noise 3.0 % and dark 0.3 %.

Read noise at 3 %. Participants routinely ask why a quieter gain setting is not used. At gain 200 the read noise is 0.9 e^- instead of 3.2, and it changes almost nothing, because read noise is already only 3.0 % of the variance. Remove it entirely and the SNR improves by $1/\sqrt{0.970} = 1.5$ %.

There is a structural reason. Choosing the sub as $3 \times$ the floor means $t = 3 \times 10R^2/B$, so $Bt = 30R^2$: the sky term is thirty times the read term *by construction*, whatever R happens to be. A quieter sensor does not lower that fraction — it just lets you reach the same fraction with shorter subs. Read noise is a *noise* term, not a *signal* term; removing noise you have already made negligible cannot add photons. Only aperture, time, throughput or a darker sky do that.

The exception is in the survey's bottom three rows, where the sub *cannot* reach $3 \times$ the floor because the guider caps it first. There the argument above does not hold, and gain 200 is the right call.

1.6 Achieved SNR and magnitude error.

$$\begin{aligned}
 N &= \left\lceil \left(\frac{10}{3.255} \right)^2 \right\rceil = \lceil 9.44 \rceil = \mathbf{10}, \quad t_{\text{total}} = 10 \times 102.4 = 1024 \text{ s} = \mathbf{17.1} \text{ min} \\
 \text{SNR}_{\text{total}} &= 3.255\sqrt{10} = \mathbf{10.29}, \quad \sigma_{\text{mag}} = \frac{1.0857}{10.29} = \mathbf{0.106} \text{ mag}
 \end{aligned}$$

1.7 Seeing. $n_{\text{pix}} \propto \text{seeing}^2$, and n_{pix} multiplies the term that dominates the budget, so the whole answer scales with the square of the seeing. On a 2 arcsec night,

$$n_{\text{pix}} = \pi (1.5 \times 0.559)^2 = 2.21 \text{ px} \quad \text{instead of} \quad 4.98 \text{ px},$$

a factor of $(2/3)^2 = 0.44$. That lifts SNR_{sub} from 3.26 to 4.66, and the observation drops from 10 subs to **5** — 8.5 minutes instead of 17.1, exactly half.

Seeing is worth 24 times more than the sensor.

Spend the effort on...	SNR gain	in magnitudes
a perfect sensor (read noise $\rightarrow 0$)	1.015×	0.016
one arcsecond of seeing ($3'' \rightarrow 2''$)	1.43×	0.39

Seeing is worth 24 times more than the sensor here, at no cost. The reason is structural: seeing sets n_{pix} , and n_{pix} multiplies the term that dominates the noise budget. This

is why site quality, focus and guiding are worth more attention than the camera spec sheet, and it is the single most transferable lesson in this assignment.

Note that it is also the second time the mount has decided the answer. Seeing as *delivered* at the detector is atmosphere plus focus plus tracking; two of those three are things this observatory controls.

Answer key — Part 1

Fastest band (L)	12.2 min — and not the answer
Chosen band	R , to join the network's existing points
Sky-limited floor	34.1 s
Saturation ceiling	5705 s (guiding cap 300 s binds instead)
Sub-exposure	102.4 s
SNR per sub	3.26
Number of subs N	10
Total on-sky time	17.1 min
σ_{mag}	0.106 mag
On a 2" night	5 subs, 8.5 min

The reply they were asked for. “Yes — 17 minutes on sky, 10 subs of 102 s in R , photometry to ± 0.11 mag.” Most estimates are much longer, because mag 20 sounds beyond a small wide-field instrument; it is not, when the aperture contains only five pixels of sky. Comparing the estimates written down at the start against the 17 minutes is worth two minutes of discussion — and comparing the *filter* guessed at the start against L is worth two more.

The second half of the reply matters as much. The star could have integrated for 5705 s before saturating, so the 102 s frame was set by the mount. On this instrument the limit on a single frame is mechanical, not photometric.

Common wrong answers, and what they mean.

- L , because it is fastest. The arithmetic is right and the answer is wrong. This is the point of 1.3; do not let it pass on the strength of a clean table.
- *Only one band surveyed.* Usually R , because the network uses R — the right answer reached without doing the work that justifies it. Half marks on 1.2, none on 1.3.
- *Ceiling ≈ 300 s.* They took the cap as the physics. Ask them what would change if the guider were perfect.
- *A narrowband floor reported as an error.* 853 s is correct. What it means is that the regime is unreachable here, not that the sum is wrong.
- $n_{\text{pix}} \approx 2.2$. They used $\pi\theta^2$ — a radius of one FWHM instead of 1.5. (Or they did the 2 arcsec case by mistake.)
- $N = 9.44$. Not rounded up. Then the achieved SNR is reported as exactly 10, which is a give-away.
- $\text{SNR}_{\text{sub}} \approx 6.3$. They forgot n_{pix} on the sky term — i.e. measured one pixel's worth of sky instead of the aperture's.
- *Read noise entered as 10.24.* R is squared in the equation; the table already gives R , not R^2 .

- *A sub-pixel FWHM reported as an error.* It is correct: at 3.58 arcsec/px the seeing disc is narrower than one pixel. See H1.2.

6.2 Part 2, what a good answer looks like

2.2 Reconciliation. The Exposure tab should return, for the Part 1 inputs: plate scale 3.58 arcsec/px, FWHM 0.84 px, aperture 5 px, sub window ≥ 34.1 s / ≤ 300.0 s, recommended sub 102.4 s, SNR/sub 3.25, 10 subs, 17.1 min total, SNR 10.3, ± 0.106 mag, *sky-limited*. An exact match is expected, because the console runs the same equations they just did by hand — that is the intended realisation.

The disagreement to look for is the optics. Selecting a saved setup with a different focal length and pixel size overwrites both fields without warning. Anyone whose total is out by a large factor should check those two boxes read 300 mm and 5.2 μ m before looking elsewhere.

2.3 Candidate selection. Accept any target that clears 30° during astronomical dark with reasonable moon separation. A good answer notes airmass at peak (below ~ 1.5 is comfortable), moon separation (a target 20° from a bright moon is a waste of a slot however high it is), and the class — an SN candidate is a better use of a follow-up node than a known AGN.

2.4 Naming the chain. All six stages, in order: slew, plate-solve, centre (the solve says where you *actually* are; the mount is corrected and re-solved), autofocus (HFR V-curve), guide (PHD2 calibrate then lock), expose $\times N$, archive. Participants who say "it slewed and then took pictures" have not watched it.

2.5 Header values.

Keyword	Expected	Where it comes from
OBJECT	the candidate's designation	the observation request
FILTER	R (or as planned)	filter wheel slot
EXPTIME	the sub length they planned	the plan, not the planner
GAIN	60	camera setting
EGAIN	$0.45 \text{ e}^-/\text{ADU}$	calibration table, <i>measured</i> , not the setting
READNOIS	3.2 e^-	calibration table
CRVAL1/2	within arcminutes of the target RA/Dec	injected from the plate solve
HFR	a few px, stable across the run	ASTAP at ingest, on the archive row
star count	tens to hundreds	ASTAP at ingest, on the archive row

The teaching point: EGAIN and READNOIS in the header are what makes the frame reducible by *somebody else*, months later, without asking you anything. That is the difference between a picture and a data product.

6.3 One-page answer card

Part	Quantity	Answer
1	plate scale / FWHM / n_{pix}	3.575 "/px / 0.839 px / 4.98 px
1	source rate at $m = 20$, R band	$1.318 \text{ e}^{-} \text{ s}^{-1}$
1	survey, broadband totals	L 12.2 / G 16.1 / R 17.1 / B 28.2 min
1	survey, narrowband totals	20.0 / 25.0 / 40.0 min
1	narrowband floors	683–1024 s, <i>above</i> the 300 s cap
1	read share, broad <i>vs.</i> narrow	3.0 % <i>vs.</i> 13–18 %
1	the choice	R — L is faster, and unusable
1	sky-limited floor (R)	34.1 s
1	saturation ceiling	5705 s (<i>guiding cap 300 s binds</i>)
1	sub-exposure	102.4 s
1	SNR per sub	3.26
1	subs / total	10 / 17.1 min
1	achieved SNR / σ_{mag}	10.29 / 0.106 mag
1	the answer to the coordinator	<i>17 minutes, in R</i>
1	on a 2 " night	5 subs / 8.5 min — exactly half
1	noise budget share (R)	sky 88.9 / source 7.8 / read 3.0 / dark 0.3 %
1	sensor <i>vs.</i> seeing	+0.016 <i>vs.</i> +0.39 mag — seeing wins 24×
2	planner, same inputs	102.4 s $\times$ 10, 17.1 min, SNR 10.3, sky-limited
2	planner, filter swept	must reproduce all four surveyed rows
2	the disagreement to look for	a saved setup overwriting 300 mm / 5.2 μm

A Doing it from the command line

The Exposure tab, the CLI and the API all call the same function. Any of these is an acceptable way to do Part 2.

```
# the whole of Part 1, from the calibration table
python -m crito.transient.exposure \
  --calibration calibration/minicam8.example.yaml \
  --mag 20.0 --snr 10 --filter R --gain 60 --temp -10 \
  --focal-length 300 --pixel-size 5.2 --seeing 3 --max-sub 300

# same thing with the constants typed in by hand (no table needed)
python -m crito.transient.exposure \
  --mag 20.0 --snr 10 --read-noise 3.2 --full-well 38000 \
  --sky 3.0 --zp 20.3 --dark 0.01 \
  --focal-length 300 --pixel-size 5.2 --seeing 3 --max-sub 300

# protect a bright field star from saturation (extra credit)
python -m crito.transient.exposure ... --brightest-mag 12

# over HTTP
curl localhost:8000/api/tools/exposure/calibration
curl -X POST localhost:8000/api/tools/exposure \
  -H 'content-type: application/json' \
  -d '{"mag":20.0,"required_snr":10,"filter":"R","gain":60,
      "temp_c":-10,"seeing_arcsec":3,"max_sub_s":300}'

# the transient chain
curl localhost:8000/api/transient/night          # tonight's dark window
curl -X POST localhost:8000/api/transient/poll    # poll ALerCE now
```

```
curl "localhost:8000/api/transient/candidates?group_by=class"
```

B Troubleshooting

Symptom	Cause / fix
focal length unknown (400)	no <code>focal_length_mm</code> in <code>observatory.yaml</code> ; pass <code>-focal-length</code> or fill the form field
calibration for gain N is incomplete filter 'X' not in calibration Exposure tab: <i>no calibration table</i>	<code>read_noise_e</code> or <code>full_well_e</code> is null in the table add it under <code>filters:</code> , or tick <i>manual constants</i> <code>CRITO_CALIBRATION_FILE</code> unset or wrong path; restart the backend
Plan wants an absurd total time <i>saturation ceiling below sky floor</i>	sky-limited faint broadband target — that <i>is</i> the answer, not a bug: check n_{pix} and the sky rate bright star in a dark sky: lower the gain (more full well) or accept a read-noise penalty
Frames are blank noise	<code>gsc</code> missing from the simulator image; nothing will plate-solve
Nothing solves	FOV hint does not match the star database — the simulated optics are 300 mm/5.2 μm , which wants D05
No <code>CRVAL1</code> / <code>CRVAL2</code> in the header	<code>CRITO_SOLVE_SCIENCE_FRAMES</code> is <code>false</code> (the default)
Queue will not launch	<i>manual override</i> is latched — a manual command always pauses the queue; clear it and resume
403 on any move command	the account is <code>sim_only</code> and this backend is not a simulator; check the <code>SIMULATION</code> badge
Candidate list empty	the broker returned nothing: lower <code>CRITO_ALERCE_PROBABILITY</code> , widen the lookback, re-poll
Polar alignment always reads ~ 0	the simulated mount is perfectly aligned; the maths is being exercised, not your judgement

C Symbols

Symbol	Meaning	Units
s	plate scale	arcsec/px
θ_{px}	seeing FWHM in pixels	px
σ	Gaussian PSF width, $\theta_{\text{px}}/2.3548$	px
n_{pix}	pixels in the photometry aperture	px
m	target magnitude	mag
ZP	zero point: the magnitude giving $1 \text{ e}^- \text{ s}^{-1}$	mag
S	source count rate	$\text{e}^- \text{ s}^{-1}$
P	PSF peak rate, central pixel	$\text{e}^- \text{ s}^{-1} \text{ px}^{-1}$
B	sky background rate	$\text{e}^- \text{ s}^{-1} \text{ px}^{-1}$
D	dark current	$\text{e}^- \text{ s}^{-1} \text{ px}^{-1}$
R	read noise (<i>not</i> squared)	e^-
FW	full well	e^-
t	sub-exposure length	s
N	number of subs	—
σ_{mag}	1- σ magnitude uncertainty	mag

Built on CRITO. The planner is `crito.transient.exposure`; the constants are `calibration/minicam8.example.yaml`. Replace them with your own measurements before you believe any of it.